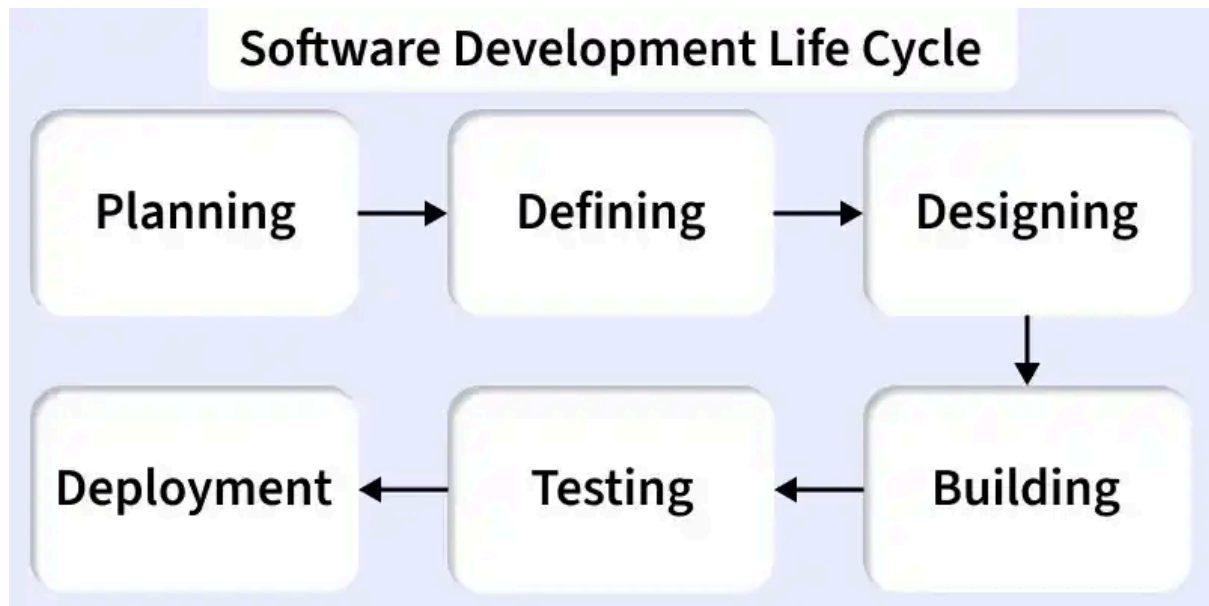


The Software Development Life Cycle (SDLC) is a structured process used to plan, design, develop, test, deploy, and maintain software. It ensures a systematic workflow and helps align software development with business goals and user requirements.

- It provides a clear and organized framework for managing the development phases.
- It helps detect errors early, reducing overall costs and time.
- Ensure the delivery of high-quality software that meets user expectations.



The stages of the software development cycle

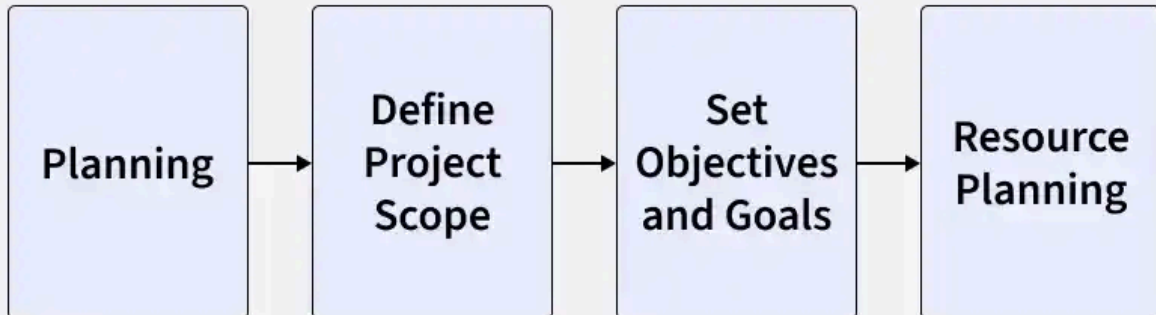
The software development lifecycle (SDLC) typically includes six or seven phases, depending on the development model used.

Phase 1: Planning and feasibility analysis

This phase determines whether the project is technically, financially, and operationally feasible.

- Activities: Feasibility analysis, cost estimation, schedule planning, and resource planning.
- Output: Project plan, Feasibility report
- Key roles: Project Management, Senior Engineers, Stakeholders

Stage-1: Planning and Requirement Analysis

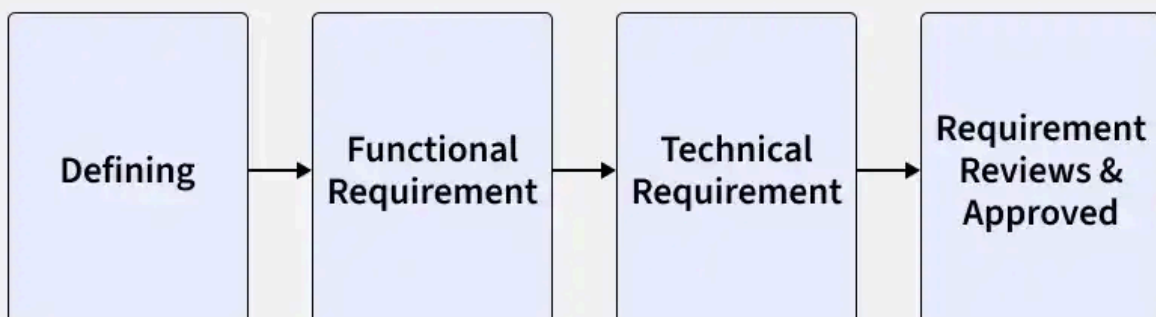


Phase 2: Requirements Specification (SRS)

At this stage, detailed functional and non-functional requirements are clearly documented and approved by stakeholders.

- Activities: Gather requirements, verify, and document them.
- Output: Software Requirements Specification (SRS)
- Key positions: Business Analyst, Product Owner

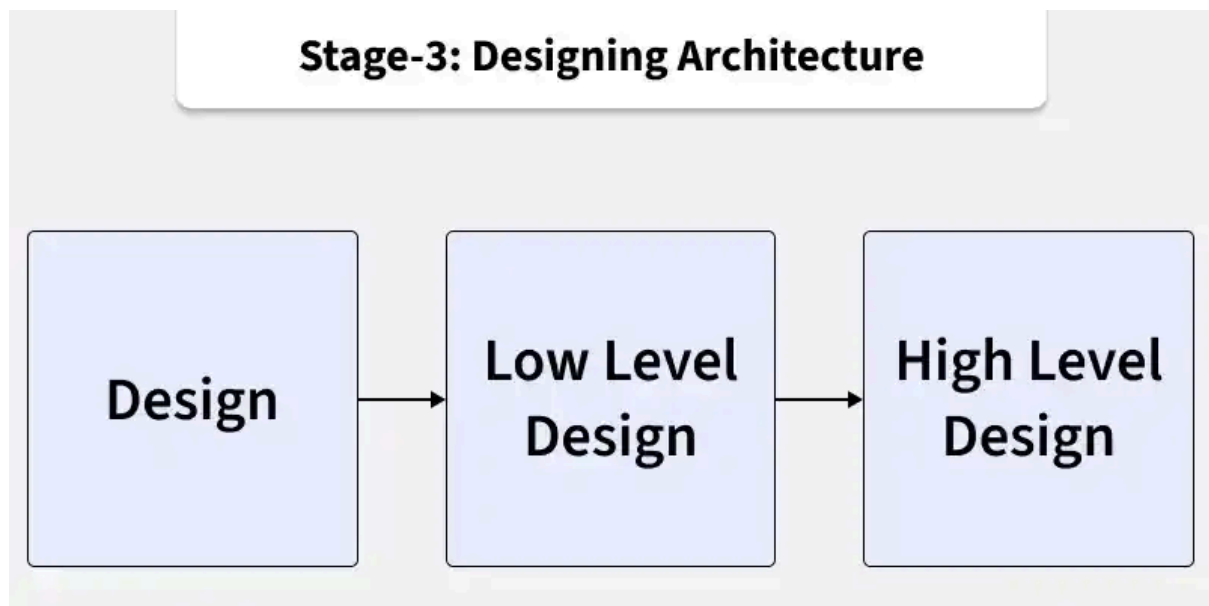
Stage-2: Defining Requirements



Phase 3: System Design

At this stage, the approved requirements will be converted into engineering designs for implementation.

- High-Level Design (HLD): Defines the system architecture, technology stack, database design, and key modules.
- Low-Level Design (LLD): Define component logic, APIs, data structures, and workflows.
- Output: Design Specification Document (DDS)

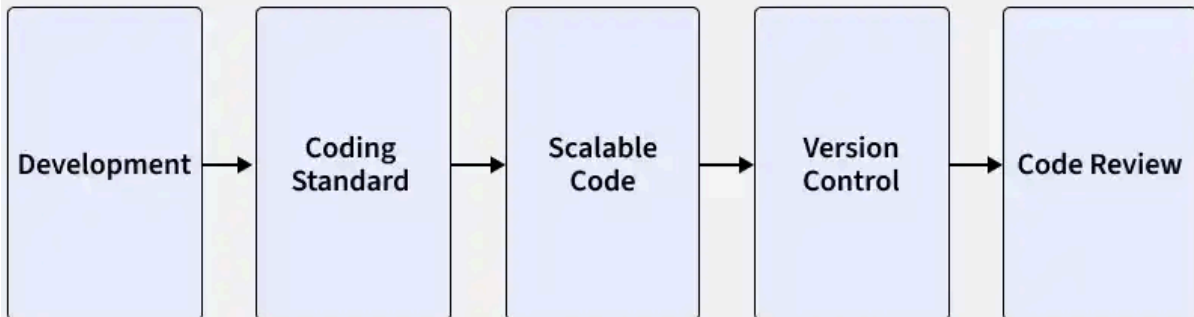


Phase 4: Development (Programming)

Software developers build software based on approved designs.

- Activities: Programming, source code review, unit testing, version management.
- Tools: Integrated development environment (IDE), version control system, debugger.
- Output: Source code, executable application
- Key positions: Programmer Frontend, Backend, Full Stack

Stage-4: Developing Product



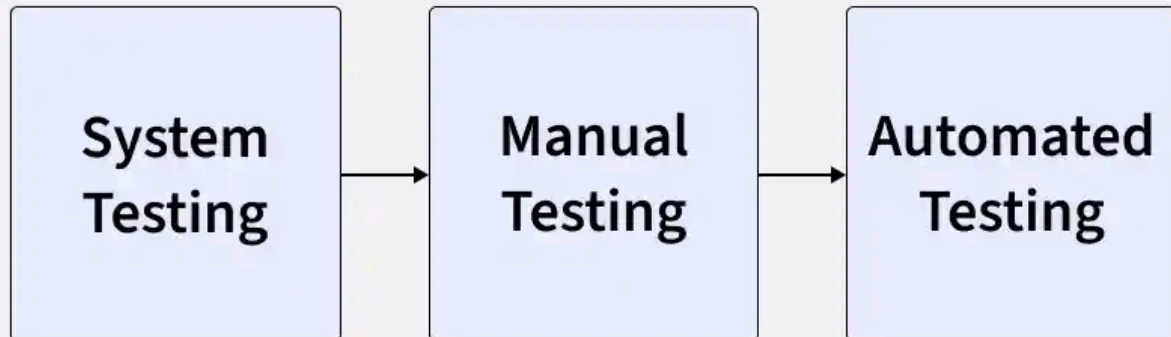
Phase 5: Testing

Testing ensures that the software meets requirements and is free of errors before release.

Types of testing include:

- Unit testing: Verify individual components
- Integration testing: Ensure the modules work together.
- System testing: Verify the entire system.
- User Acceptance Testing (UAT): Confirm that business requirements have been met.
- Output: Test cases, bug reports, quality metrics.

Stage-5: Product Testing and Integration

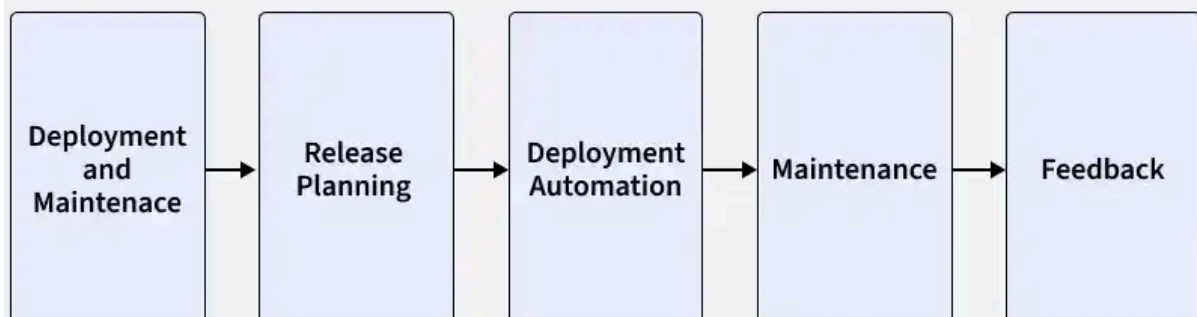


Phase 6: Implementation

The software has been tested and released to users.

- Activities: Production setup, deployment, and preliminary testing.
- Modern methods: Continuous Integration and Continuous Deployment (CI/CD) workflows enable faster and more reliable releases.
- Result: The application is running.
- Key positions: DevOps Engineer, Software Release Manager

Stage 6: Deployment and Maintenance of Products



Phase 7: Maintenance

Post-deployment support ensures long-term usability.

- Activities: Bug fixes, performance optimization, updates, and feature upgrades.
- Output: Patches, updates, new versions
- Key positions: Support Engineer, Programmer